

RESEARCH PAPER

Temperature fine-tunes Mediterranean *Arabidopsis thaliana* life-cycle phenology geographically

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ABSTRACT

- To understand how adaptive evolution in life-cycle phenology operates in plants, we need to unravel the effects of geographic variation in putative agents of natural selection on life-cycle phenology by considering all key developmental transitions and their co-variation patterns. We address this goal by quantifying the temperature-driven and geographically varying relationship between seed dormancy and flowering time in the annual *Arabidopsis thaliana* across the Iberian Peninsula.
- We used data on genetic variation in two major life-cycle traits, seed dormancy (DSDS50) and flowering time (FT), in a collection of 300 *A. thaliana* accessions from the Iberian Peninsula. The geographically varying relationship between life-cycle traits and minimum temperature, a major driver of variation in DSDS50 and FT, was explored with geographically weighted regressions (GWR). The environmentally varying correlation between DSDS50 and FT was analysed by means of sliding window analysis across a minimum temperature gradient.
- Maximum local adjustments between minimum temperature and life-cycle traits were obtained in the southwest Iberian Peninsula, an area with the highest minimum temperatures. In contrast, in off-southwest locations, the effects of minimum temperature on DSDS50 were rather constant across the region, whereas those of minimum temperature on FT were more variable, with peaks of strong local adjustments of GWR models in central and northwest Spain. Sliding window analysis identified a minimum temperature turning point in the relationship between DSDS50 and FT around a minimum temperature of 7.2 °C. Above this minimum temperature turning point, the variation in the FT/DSDS50 ratio became rapidly constrained and the negative correlation between FT and DSDS50 did not increase any further with increasing minimum temperatures.
- The southwest Iberian Peninsula emerges as an area where variation in life-cycle phenology appears to be restricted by the duration and severity of the hot summer drought. The temperature-driven varying relationship between DSDS50 and FT detected environmental boundaries for the co-evolution between FT and DSDS50 in *A. thaliana*. In the context of global warming, we conclude that *A. thaliana* phenology from the southwest Iberian Peninsula, determined by early flowering and deep seed dormancy, might become the most common life-cycle phenotype for this annual plant in the region.

INTRODUCTION

Pronounced environmental heterogeneity in soil fertility, climate seasonality and fire frequency typically characterises Mediterranean climate regions. Broad agreement exists that environmental heterogeneity chiefly accounts for the high regional plant diversity observed in these scarce climate

regions, *i.e.* almost 20% of known vascular plant species occur in less than 5% of the Earth's surface (Cowling *et al.* 1996; Lobo *et al.* 2001). In addition, the Quaternary glacial history of some Mediterranean climate regions, *e.g.* the Mediterranean Basin, has also played a very important role in structuring their current patterns of plant biodiversity (Comes & Kadereit 1998; Hewitt 1999; Hughes *et al.* 2006; Médail & Diadema 2009). As

a result of such inherent environmental heterogeneity, the agents of natural selection are assumed to have been acting at a fine scale, which is also considered an additional explanation for the high plant diversity of these regions (Cowling *et al.* 1996). Thus, Mediterranean climate regions are appropriate natural scenarios in which to undertake evolutionary studies on the processes underlying plant differentiation.

One of these evolutionary processes deals with the adaptive adjustment of life-cycle phenology, *i.e.* seasonal timing of developmental events spanning from seed to seed, to heterogeneous environments (Rathcke & Lacey 1985; Donohue 2005; Wilczek *et al.* 2009; Huang *et al.* 2010; Chiang *et al.* 2013; Burghardt *et al.* 2015). Life-cycle phenology must be regarded as a composite trait influenced by the interaction between genetic and environmental factors and whose adaptive variation enhances long-term population fitness. Variation in the life-cycle phenology components with the highest impact on fitness, *i.e.* timing of seed germination and timing of reproduction, are expected to be related to the variation exhibited through their putative agents of natural selection (Caicedo *et al.* 2004; Stinchcombe *et al.* 2004; Lempe *et al.* 2005; Shindo *et al.* 2005; Méndez-Vigo *et al.* 2011; Kronholm *et al.* 2012; Manzano-Piedras *et al.* 2014; Vidigal *et al.* 2016). As climate broadly acts as a major selective pressure on life-cycle phenology, adaptive patterns of variation in continuous populations are commonly captured through distinctive geographic clines, revealing the interplay between life-cycle phenology, genetic loci that govern fitness-related life-cycle traits and climate factors (Aranzana *et al.* 2005; Atwell *et al.* 2010; Fournier-Level *et al.* 2011, 2013; Hancock *et al.* 2011; Méndez-Vigo *et al.* 2011; Savolainen *et al.* 2013; Vidigal *et al.* 2016).

It is becoming increasingly accepted that the evolutionary implications of variation in life-cycle phenology cannot be fully understood if its components are considered in isolation (Toorop *et al.* 2012; Debieu *et al.* 2013; Burghardt *et al.* 2016). As a matter of fact, the success of an organism when facing heterogeneous or novel environments depends on its ability to coordinate adaptive shifts in fitness-related traits that are closely correlated with each other (Montesinos-Navarro *et al.* 2011; Fischer *et al.* 2016). Recent studies on the annual plant *Arabidopsis thaliana* indeed showed the influence that flowering time, which strongly determines seed maturation conditions and post-dispersal temperatures, has on seed dormancy induction and germination behaviour (Donohue 2005, 2009, 2014; Finch-Savage & Leubner-Metzger 2006; Chiang *et al.* 2011; Burghardt *et al.* 2015, 2016; Springthorpe & Penfield 2015). Hence, seed dormancy and flowering time, given their impact on fitness, are expected to be tightly correlated, and will eventually determine adaptive variation in life-cycle phenology.

Such a correlation has recently been found in *A. thaliana*, indicating that seed dormancy and flowering time are negatively correlated at both regional (Vidigal *et al.* 2016) and global (Debieu *et al.* 2013) scales. Interestingly, patterns of co-variation between seed dormancy and flowering time appear to vary geographically and independently of major genetic population structure. This supports the view that geographic patterns of variation in life-cycle phenology are adaptive (Chuine 2010; Debieu *et al.* 2013) and that correlations between fitness-related traits need not be constant over space, and probably over time (Fischer *et al.* 2016). Nevertheless, gaining an in-depth understanding of the evolutionary potential of co-variation between

seed dormancy and flowering time might be more complicated than expected. For example, Debieu *et al.* (2013) used a set of 112 *A. thaliana* accessions to show how vegetative growth, flowering time and seed dormancy followed a latitudinal cline across Europe, and how co-variation patterns between these traits differed between North and South Europe. However, only latitude appeared to have some explanatory power in their study, explaining only up to 5% and 11% of the observed variation in seed dormancy and flowering time, respectively (Debieu *et al.* 2013). In contrast, Vidigal *et al.* (2016), using a collection of 300 *A. thaliana* accessions exclusively from the Iberian Peninsula, observed that early-flowering accessions with very high seed dormancy mostly occurred in the southwest Iberian Peninsula. In this case, the high number of geo-referenced accessions allowed identification of minimum temperature as a major driver of variation in these traits. Average annual minimum temperature alone explained up to 44.8% and 39.5% of the genetic variation in seed dormancy and flowering time, respectively (Vidigal *et al.* 2016). Overall, these results stress the value of high-density regional collections of natural accessions to study large-scale variation in the life-cycle phenology of *A. thaliana*. However, regional patterns can vary substantially across the species' distribution range, which eventually determine the density of populations needed to detect such patterns.

The main goal of this study is to reveal and quantify the temperature-driven and geographically varying relationship between seed dormancy and flowering time in *A. thaliana* across the Iberian Peninsula in the west Mediterranean Basin, which is the distribution area of *A. thaliana* known to harbour the largest regional genomic diversity (Picó *et al.* 2008; Cao *et al.* 2011; Weigel 2012; Brennan *et al.* 2014; The 1001 Genomes Consortium 2016). We use recent data on genetic variation in seed dormancy and flowering time for the 300 Iberian *A. thaliana* accessions mentioned above (Vidigal *et al.* 2016). We hypothesise that the negative correlation exhibited between seed dormancy and flowering time is spatially heterogeneous across the Iberian Peninsula. It is already known that temperature strongly shapes phenotypic variation in Iberian *A. thaliana*, with warmer environments selecting for early flowering (Méndez-Vigo *et al.* 2011; Manzano-Piedras *et al.* 2014) and high seed dormancy (Vidigal *et al.* 2016). Beyond this general trend, however, we ignore how the species fine-tunes life-cycle phenology across the region, since temperature sharply changes along altitudinal and latitudinal clines. This requires a spatially explicit approach to quantify the effects of temperature variation, not only on seed dormancy and flowering time separately, but on the relationship between these two major components. This knowledge will help better understand how adaptive evolution in life-cycle phenology operates in plants (Debieu *et al.* 2013), as well as inferring how plants will manage to keep up with global climate change by modifying their life-cycle phenology (Wilczek *et al.* 2014).

MATERIAL AND METHODS

Source accessions

The 300 Iberian *A. thaliana* accessions are part of a long-term project to generate a permanent collection of natural populations from the southwest Mediterranean Basin across Spain, Portugal and Morocco for the integrative and comprehensive

study of the species' evolutionary ecology and functional genetics (Picó *et al.* 2008; Méndez-Vigo *et al.* 2011; Brennan *et al.* 2014; Manzano-Piedras *et al.* 2014; Marcer *et al.* 2016; Vidigal *et al.* 2016). For this study, we surveyed natural populations across the entire Iberian Peninsula (36.52° N–43.40° N, 3.19° E–8.54° W; 1–2662 m a.s.l.; between-population distance 1–1042 km) continuously between 2000 and 2010. We collected seed from several individuals per population, depending on population size. A few months after the field season of every sampling year, individuals were multiplied using the single seed descent method in controlled conditions in glasshouses at the Centro Nacional de Biotecnología (Madrid, Spain) and stored in dry conditions in cellophane bags at room temperature in darkness. When possible, accessions were chosen based on their flowering time and/or vernalisation requirement during the multiplication experiments. This is important for our study as we selected one individual with a common phenotype within its population, increasing the odds of using those phenotypes best suited to their local environments. By selecting a common phenotype per population, we are also buffering the effects, if any, of changing environmental conditions during the sampling period (2000–2010) on life-cycle traits. Here, we assume that temporal changes in environmental conditions during this period of time have not substantially affected population mean values of major life-cycle traits of interest, as suggested from preliminary data (F.X. Picó, unpublished observations). Study accessions were genetically different from each other based on 250 polymorphic genome-wide neutral single nucleotide polymorphisms (SNP; Manzano-Piedras *et al.* 2014). In 2013, all accessions were multiplied again in the greenhouses at Wageningen University (Wageningen, the Netherlands) to synchronise their seed production, which is essential to obtain accurate seed dormancy estimates (Vidigal *et al.* 2016).

Life-cycle phenology

The life-cycle phenology of *A. thaliana* was characterised by variation in two major life-cycle traits: seed dormancy and flowering time. For each accession, seed dormancy was quantified as DSDS50, *i.e.* days of seed dry storage required to reach 50% germination (see He *et al.* 2014 for details on calculation), whereas flowering time (FT hereafter) was quantified as the number of days from planting date to anthesis of the first flower. To estimate DSDS50, vernalised mother plants were first grown in standard conditions in a glasshouse (20 °C day/18 °C night, 16-h light:8-h dark, 70% relative humidity) to synchronise seed harvest among all accessions. Subsequently, seed was used to estimate DSDS50 in an incubator at 22 °C and constant light for 5 days (see Vidigal *et al.* 2016). FT was estimated in a growth chamber at 21 °C with a long-day photoperiod (16-h light:8-h dark; see Méndez-Vigo *et al.* 2011). Experiments to estimate DSDS50 and FT lasted for a maximum of 559 and 220 days, respectively (Méndez-Vigo *et al.* 2011; Vidigal *et al.* 2016). Both seed dormancy and FT were estimated under controlled laboratory conditions, thus providing the quantitative genetic component for both traits.

Minimum temperature data

Previous studies indicated that minimum temperature is the main contributing factor to the observed patterns of variation

in DSDS50 and FT in Iberian *A. thaliana*, far exceeding other environmental factors (Méndez-Vigo *et al.* 2011; Manzano-Piedras *et al.* 2014; Vidigal *et al.* 2016). Given the predominant role of minimum temperature in shaping patterns of phenotypic variation in Iberian *A. thaliana*, for the sake of clarity and simplicity we used the average annual minimum temperature (minimum temperature hereafter) as explanatory variable in all analyses. Additional analyses using variables related to precipitation, vegetation and soil, alone or in combination with minimum temperature, neither improved the outcomes nor added any value to interpretation of the resulting patterns in a significant way (results not shown). Minimum temperature from every population was obtained from the Digital Climatic Atlas of the Iberian Peninsula (Ninyerola *et al.* 2000). Minimum temperature data were from a spatial interpolation using temperature records from a total of 3608 meteorological stations across Portugal and Spain during the period 1950–1999. Hence, although our *A. thaliana* populations were sampled between 2000 and 2010, the minimum temperature data used in this study reflect environmental conditions where *A. thaliana* has probably been occurring and adapting for a long time in the Iberian Peninsula.

Statistical analyses

The hypothesised geographically varying relationship between life-cycle phenology traits and minimum temperature was explored with geographically weighted regression (GWR) models (Fotheringham *et al.* 2002) using SAM software (Rangel *et al.* 2010). First, we performed a GWR to test the effect of minimum temperature on DSDS50 and another GWR to test the effect of minimum temperature on FT. GWR is a spatially explicit explanatory regression where regression parameters are estimated at each sampled location *i* as follows:

$$y_i = \beta(v_i, v_i)x_i + \varepsilon_i$$

where y_i is the dependent variable, x_i is the independent variable, ε_i is the Gaussian error, (v_i, v_i) are the geographic coordinates, and $\beta(v_i, v_i)$ is the GWR varying coefficient at each location *i*. GWR performed a local regression at each of the 300 locations by taking a fraction (10–15%) of the neighbouring locations into account using an optimal bandwidth size. For each location, the selected bandwidth was optimised using a Gaussian spatial weighting function minimising the Akaike information criterion (AICc). This procedure generally enhances the goodness-of-fit of GWR by setting the optimal number of neighbouring locations to perform local regressions, as well as by controlling for border and sampling effects. GWR and OLS (ordinary least squares) results were also compared with an ANOVA to check that the explanatory power of GWR was better than that of OLS. The lack of autocorrelation of the residuals was checked with a spatial correlogram plotting Moran's *I* values of GWR residuals. Thus, GWR provided local estimates of the adjustment of DSDS50 and FT values to minimum temperatures in Iberian *A. thaliana* populations, *i.e.* local adjusted pseudo- R^2 values.

After assessing the relationship between minimum temperature and DSDS50 and FT in two separate GWR models, we uncovered the environmentally varying correlation between DSDS50 and FT by means of sliding window analysis (as in

Debieu *et al.* 2013). Sliding window analyses are used to display the dynamic relationship between variables across a continuous space. In this study, we used a temperature sliding window to account for the co-variation between life-cycle traits across a minimum temperature gradient. We selected a temperature sliding window instead of a geographic sliding window because minimum temperature is better than geography in characterising the environment of Iberian *A. thaliana* populations. For example, geographically close populations can exhibit pronounced differences in minimum temperature if they are located along the same altitudinal gradient (Montesinos *et al.* 2009; Méndez-Vigo *et al.* 2011). Hence, all 300 *A. thaliana* populations were first ranked according to their minimum temperatures (range = -0.7 – 13.0 °C). For every population along the minimum temperature gradient and regardless of its geographic location, we sub-sampled all populations whose minimum temperatures fell within a temperature interval around it. We tested different temperature breadths and eventually selected a temperature interval of 7 °C, centred on the minimum temperature characterising each population. The temperature breadth determines the number of populations falling within the temperature interval as well as the truncation of the sampling window for those populations at the low and high ends of the temperature gradient. Consistent patterns were detected with narrower temperature breadths (results not shown) but the chosen one optimised the patterns observed.

For each of the 300 values of the minimum temperature gradient, we computed the correlation between DSDS50 and FT as well as the mean and SD of the FT/DSDS50 ratio using all the accessions falling within the temperature interval of 7 °C (mean $\pm$ SD number of sampling accessions per minimum temperature = 194.6 ± 53.1). Note that we used more than 100 accessions in 93% of the analyses (279 of 300 analyses). The FT/DSDS50 ratio provided an integrated measure of variation between the two life-cycle traits because accessions with early flowering times and high seed dormancy had low FT/DSDS50 ratios, whereas late flowering accessions with low seed dormancy exhibited high FT/DSDS50 ratios. The significance of

the correlations between DSDS50 and FT for every minimum temperature was tested with the Dutilleul's modified *t*-test, using the R package SpatialPack (Osorio & Vallejos 2014), to account for any spatial autocorrelation in the data. The Dutilleul's modified *t*-test corrects the variance of the test statistic and the degrees of freedom according to the extent of spatial autocorrelation of each variable of the correlation (Dutilleul 1993). Our large sample size ($N = 300$) ensures high accuracy of the results, as required in spatially explicit statistical tests (Legendre *et al.* 2002).

RESULTS

The geographically varying relationship between life-cycle phenology traits, *i.e.* DSDS50 and FT, and minimum temperature was explored with two separate geographically weighted regression (GWR) models. For DSDS50 and minimum temperature, the GWR results showed that local models based on GWR performed better than the global model based on the classical ordinary least squares (OLS) regression model ($F_{5,9,292.1} = 6.68$, $P < 0.0001$; ANOVA; adjusted $R^2 = 0.41$ and 0.34 for GWR and OLS regression, respectively). All local models were significant ($P < 0.001$ in all cases) and all local standardised coefficients of regression were positive (range $r = 0.33$ – 0.90), indicating that minimum temperature and DSDS50 were positively correlated in all cases, *i.e.* accessions from warmer locations exhibited higher seed dormancy. The local adjusted pseudo- R^2 values (range of pseudo- $R^2 = 0.18$ – 0.50) clearly increased along a north–south latitudinal gradient, reaching the highest values in the southwest Iberian Peninsula (Fig. 1A).

For FT and minimum temperature, the GWR results also showed that local models performed better than the global OLS model ($F_{9,4,288.6} = 8.90$, $P < 0.0001$; ANOVA; adjusted $R^2 = 0.49$ and 0.36 for GWR and OLS regression, respectively). All local models were significant ($P < 0.003$ in all cases) and all local standardised coefficients of regression were negative (range of $r = -0.31$ to -1.00). Therefore, minimum temperature and FT were negatively correlated in all cases, *i.e.* accessions from

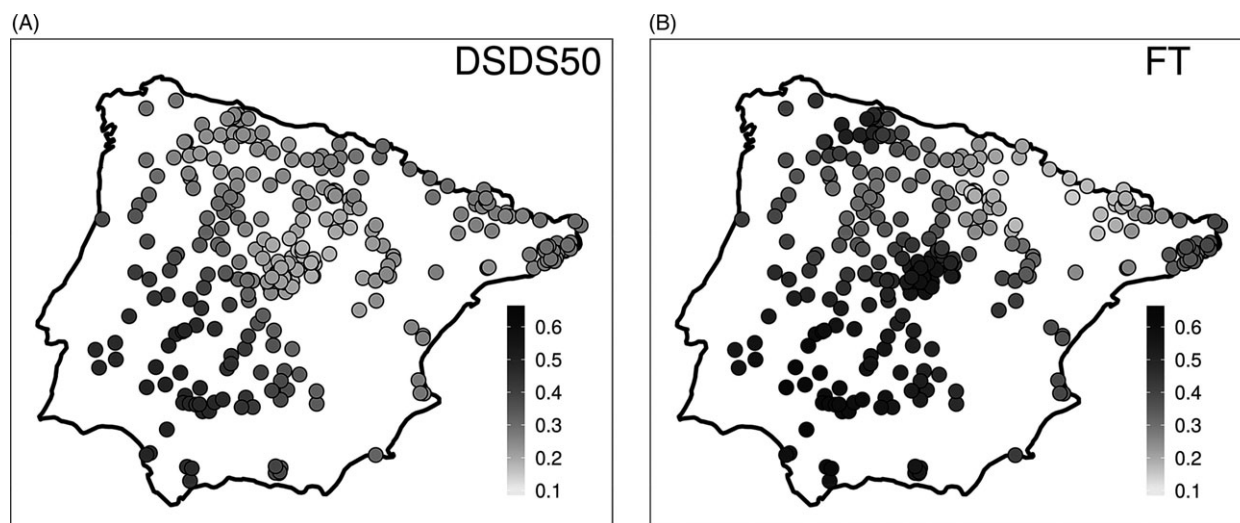


Fig. 1. A: Map showing the geographic position of accessions indicating their GWR local adjusted pseudo- R^2 values for DSDS50 and minimum temperature. B: Map showing the geographic position of accessions indicating their GWR local adjusted pseudo- R^2 values for FT and minimum temperature.

warmer locations flowered earlier. The local adjusted pseudo- R^2 values for FT were more variable (range pseudo- $R^2 = 0.13$ – 0.65) and exhibited more geographic complexity than those estimated for DSDS50 (Fig. 1B). In particular, the southwest Iberian Peninsula was again an area where adjustment between minimum temperature and FT peaked, but strong adjustment peaks in other nuclei of accessions in central and northwest Spain were also detected (Fig. 1B).

The environmentally varying relationship between DSDS50 and FT was analysed with a temperature sliding window analysis. The results depicted co-variation between DSDS50 and FT along a minimum temperature gradient. The mean ($\pm$ SD) of the FT/DSDS50 ratio computed from all accessions falling within the temperature interval of 7°C along the minimum temperature gradient varied from a low of 0.26 ± 0.21 , *i.e.* accessions with early flowering time and high seed dormancy, to a high of 2.10 ± 0.79 , *i.e.* accessions with late flowering time and low seed dormancy (Fig. 2A). As expected, the mean of the FT/DSDS50 ratio progressively decreased with increasing minimum temperatures (Fig. 2A). In contrast, the slope of the relationship between minimum temperature and the SD of the FT/DSDS50 ratio exhibited a turning point at a minimum temperature around 7.2°C : SD of the FT/DSDS50 ratio decreased faster with minimum temperatures above this turning point (Fig. 2B).

The temperature sliding window analysis also indicated that the correlation between FT and DSDS50 was not significant for populations with minimum temperatures $<3.6^\circ\text{C}$ and $>12.3^\circ\text{C}$ ($N = 42$ out of 300; Fig. 2C). In between these two thresholds, the correlation between FT and DSDS50 along the minimum temperature axis was significant and negative, but progressively increased with increasing minimum temperatures around 7.3°C (range of r values before the turning point = -0.18 to -0.39 ; Fig. 2C). From that minimum temperature onwards, the significant correlation between FT and DSDS50 first had relatively constant coefficients close to the maximum observed value, then ended with more variable behaviour with increasing minimum temperature (range of r

values after the turning point = -0.30 to -0.41 ; Fig. 2C). It is worth noting that the turning point in the relationship between FT and DSDS50 along a minimum temperature gradient was also observed with additional sliding window analyses depicting the correlation between each life-cycle trait and minimum temperature along the minimum temperature gradient (Fig. S1). These results showed that the correlation between DSDS50 and minimum temperature and between FT and minimum temperature also reached a major turning point very close to those reported above.

The representation of two groups of populations, whose minimum temperature is above and below the turning point for the correlation between FT and DSDS50, onto a minimum temperature map provides hints on the geographic and ecological differentiation of *A. thaliana* governed by minimum temperature (Fig. 3). In particular, *A. thaliana* populations with minimum temperatures below the turning point are mostly located in the northern half of the Iberian Peninsula, including the main mountain ranges (*i.e.* Pyrenees, Cantabrian, Central and Iberian ranges), the northern plateau and the Sierra Nevada in south Spain with the highest mountain of the Iberian Peninsula (Fig. 3). In contrast, *A. thaliana* populations with minimum temperatures above the turning point occur in the southern half of the Iberian Peninsula as well as in coastal locations around the region (Fig. 3). As a result, the two groups of *A. thaliana* populations occurring below ($N = 191$) and above ($N = 109$) the minimum temperature turning point for the correlation between FT and DSDS50 significantly differed in altitude (mean $\pm$ SE = 973.7 ± 24.0 and 459.4 ± 24.1 m a.s.l. for the former and the latter, respectively; $F_{1,298} = 198.61$, $P < 0.0001$; one-way ANOVA). In addition, higher elevation populations with minimum temperatures below the turning point also exhibited a higher percentage of wild habitat than lower elevation populations with minimum temperatures above the turning point (mean $\pm$ SE percentages of wild habitat = $64.4 \pm 2.6\%$ and $45.3 \pm 3.7\%$ for the former and the latter, respectively; $F_{1,298} = 18.95$, $P < 0.0001$; one-way ANOVA). This makes sense because human activity having more

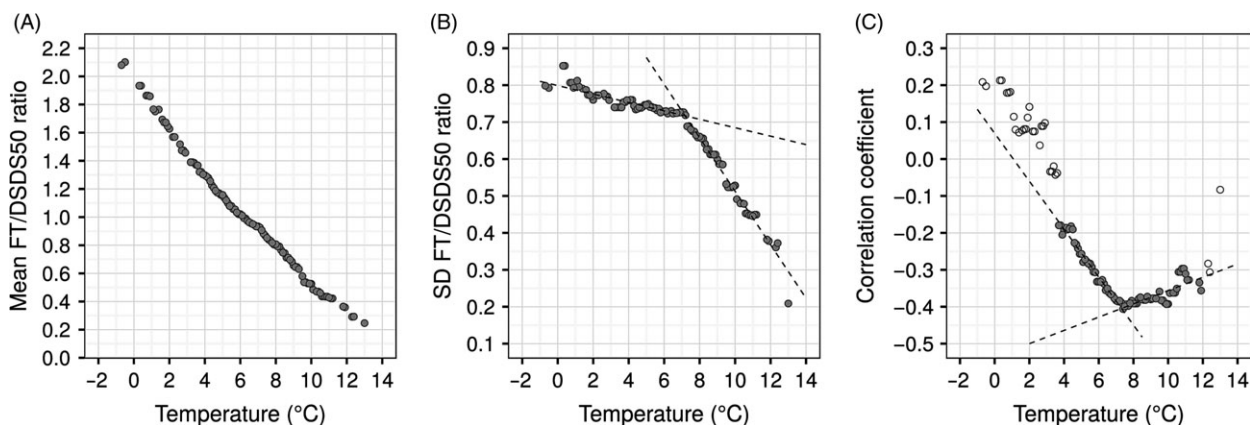


Fig. 2. A: Relationship between minimum temperature and mean of the FT/DSDS50 ratio. B: Relationship between minimum temperature and SD of the FT/DSDS50 ratio. Regression lines are given for accessions below and above the turning point at 7.2°C , which maximises the R^2 values of the two groups of accessions. C: Relationship between minimum temperature and correlation coefficients between FT and DSDS50. Filled dots indicated significant correlation coefficients based on Dutilleul's modified t -test. Regression lines are given for accessions with significant correlation coefficients below and above the turning point at 7.3°C , which maximises the R^2 values of the two groups of accessions. In all three cases, values were obtained from temperature sliding window analyses and ranked along the minimum temperature gradient.

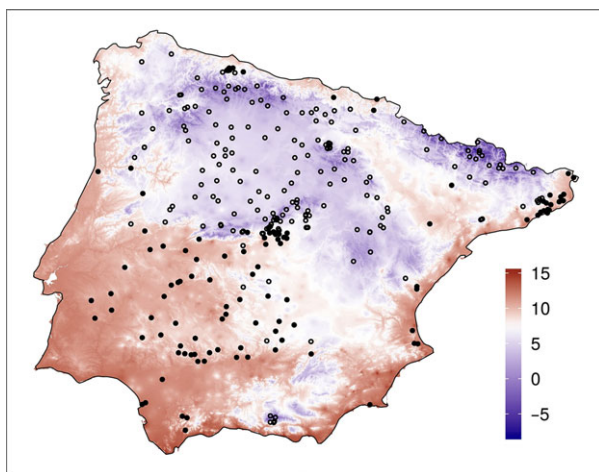


Fig. 3. Minimum temperature map for the Iberian Peninsula indicating the geographic position of *A. thaliana* populations with minimum temperatures below (hollow dots) and above (filled dots) the turning point of 7.3 °C for the correlation between FT and DSDS50 along the minimum temperature gradient (see Fig. 2C).

dramatic impacts on natural plant communities has traditionally been concentrated at lower elevations. In contrast, the two groups of populations did not differ in total annual precipitation, mean annual solar radiation or soil pH ($P > 0.10$ in all cases; one-way ANOVA). Overall, these results stress the explanatory power of minimum temperature for variation in life-cycle phenology in Iberian *A. thaliana*.

DISCUSSION

Mediterranean climate regions represent outstanding natural scenarios in which to address a multitude of evolutionary issues in plants. Adaptive variation in life-cycle phenology is particularly appropriate because major events of life-cycle phenology are markedly influenced by climate factors, whose seasonality, in fact, defines the Mediterranean climate. It is clear that we need to unravel the effects of climate factors, not only on life-cycle traits separately, but on life-cycle phenology as a whole (Post *et al.* 2008; Debieu *et al.* 2013; Burghardt *et al.* 2016). This is supported by increasingly abundant studies on *A. thaliana* as an evolutionary model system, focused on the tight interplay between flowering time and seed germination (Donohue 2005, 2009, 2014; Finch-Savage & Leubner-Metzger 2006; Chiang *et al.* 2011; Toorop *et al.* 2012; Debieu *et al.* 2013; Wilczek *et al.* 2014; Burghardt *et al.* 2015, 2016; Springthorpe & Penfield 2015), as well as on the effects of seasonal environmental variation on key developmental timing genes affecting multiple developmental transitions (Chiang *et al.* 2009; Atwell *et al.* 2010; Chen *et al.* 2014; Huo *et al.* 2016). Nevertheless, the separate analyses of the geographically varying relationship between minimum temperature and life-cycle traits of *A. thaliana* still provide valuable elements that are worth considering. For example, the GWR results indicated that the southwest Iberian Peninsula represents an area with strong local adjustments in both FT and DSDS50 (Fig. 1). This confirms the critical role of minimum temperatures in shaping life-cycle phenology towards early flowering time and strong seed dormancy as the environment becomes warmer (Méndez-

Vigo *et al.* 2011; Manzano-Piedras *et al.* 2014; Vidigal *et al.* 2016).

Interestingly, our analyses indicated that for the rest of the Iberian Peninsula, there were large differences in the geographically varying effect of minimum temperature on each of the studied life-cycle traits. On the one hand, the positive relationship between minimum temperature and DSDS50 remained rather moderate and homogeneous for accessions from off-southwest locations (Fig. 1A). Overall, these results suggest that minimum temperature increases its intensity as an agent of natural selection for DSDS50 towards the southwest Iberian Peninsula, *i.e.* an area with the highest minimum temperatures. Based on this, we hypothesise that the windows of opportunity for germination also follow a north-south gradient, becoming narrower towards the southwest Iberian Peninsula as a result of the longer and more severe summer droughts in that area. Although field demographic studies from northern *A. thaliana* populations indicated that germination peaks in early/mid-autumn, and to a lesser extent between late winter and early spring (Montesinos *et al.* 2009; Picó 2012), we still lack germination data from southern populations. Therefore, further fieldwork is needed to quantify in detail the geographic patterns of variation in seed germination behaviour in natural *A. thaliana* populations across the warmest area of the Iberian Peninsula to test this hypothesis.

On the other hand, the negative relationship between minimum temperature and FT in off-southwest locations was far more variable, with additional areas of pronounced local adjustments of the GWR models, especially in central and northwest Spain (Fig. 1B). In this case, orography might explain these results because central and northwest Spain are mountainous areas with remarkable altitudinal gradients, which imply rapid change in ecological conditions over relatively short distances. In fact, altitudinal gradients represent an important source of phenotypic and adaptive variation in *A. thaliana* in different regions across its distribution range (Montesinos *et al.* 2009; Méndez-Vigo *et al.* 2011; Montesinos-Navarro *et al.* 2011; Picó 2012; Suter *et al.* 2014; Luo *et al.* 2015a,b; Tyagi *et al.* 2016; but see Günther *et al.* 2016). We have previously shown that decreasing temperatures with increasing altitude in Iberian mountains are related to late flowering in *A. thaliana* through the obligate vernalisation requirement (Méndez-Vigo *et al.* 2011). This is a flowering-inductive effect of low temperature that appears to be restricted to altitudes >800 m (Méndez-Vigo *et al.* 2011), locations with typically cooler climates that are practically absent in the southwest Iberian Peninsula (Marcer *et al.* 2016). Hence, the preponderance of a vernalisation requirement to trigger flowering in high altitude, cool environments and the lack thereof in generally warmer lowland environments illustrates the variety and complexity of the genetic mechanisms underlying adaptive variation in FT in *A. thaliana* (Koornneef *et al.* 1998; Simpson & Dean 2002; Alonso-Blanco *et al.* 2009; Wilczek *et al.* 2009).

On top of these geographically varying relationships between minimum temperature and life-cycle traits, in this study we took a step further towards understanding how the relationship between FT and DSDS50 varied across a minimum temperature gradient. The results of the temperature sliding window analysis indicated that FT and DSDS50 co-varied negatively with minimum temperature, in agreement with the behaviour of each life-cycle trait with minimum temperature shown in

this study, as well as with previous observations (Debieu *et al.* 2013; Vidigal *et al.* 2016). The novel contribution of the sliding temperature analysis is that it allowed visualisation and quantification of the heterogeneity of the relationship between FT and DSDS50 along the minimum temperature gradient. Of particular relevance is the existence of a minimum temperature turning point in the relationship between FT and DSDS50, which has interesting ecological and evolutionary implications to better understand life-cycle phenology, not only for *A. thaliana*, but for several other annuals and short-lived perennials strongly relying on fast life cycles to succeed in their environments (García *et al.* 2008).

For example, *A. thaliana* populations with minimum temperatures above the turning point exhibited more rapid decreasing variability in the FT/DSDS50 ratio with respect to increasing minimum temperature (Fig. 2B). This means that variation in life-cycle phenology of *A. thaliana* becomes progressively constrained as minimum temperatures become higher. In Iberian warm environments, the duration and severity of the summer drought force *A. thaliana* to complete its life cycle quickly, which can only be accomplished by becoming dormant during the long hot summer, together with early flowering before temperatures become too high again. In fact, *A. thaliana* populations from the southwest Iberian Peninsula shed seed between mid-April and early May and, in the warmest locations, between late February and mid-March (C. Alonso-Blanco and F.X. Picó, personal observation). Such early flowering and fruiting imply that the maternal and post-dispersal environments encountered by seeds might also promote deeper physiological seed dormancy, ensuring late germination when the summer drought is over and arrival of the first autumn rains.

Another implication of the existence of a minimum temperature turning point in the relationship between FT and DSDS50 involves the strength of the correlation between the two life-cycle traits. The turning point identified a minimum temperature threshold beyond which the correlation coefficient between FT and DSDS50 cannot increase any further (Fig. 2C). Hence, the temperature sliding window analysis provides a tool to detect environmental boundaries for the co-evolution between FT and DSDS50 for *A. thaliana* in Mediterranean environments. In particular, below the minimum temperature threshold, *A. thaliana* shows substantial phenotypic space to adjust its life-cycle phenology by progressively advancing flowering time and increasing seed dormancy with increasing minimum temperature. Above the minimum temperature threshold, *A. thaliana* might have reached the limit for adjusting its life-cycle phenology described here, which otherwise does not prevent the species from succeeding in warmer environments. Interestingly, populations characterised by minimum temperatures above the turning point occur in the southern Iberian Peninsula and in warmer coastal areas all over the Iberian Peninsula, whereas those below the turning point mostly occur in the northern mountainous Iberian Peninsula (Fig. 3). Thus, unravelling the phenotypic limits for co-variation between key life-cycle traits and circumscribing such limits into a geographic and ecological space, in our case strongly characterised by variation in minimum temperature, might be more enlightening than depicting patterns of variation along geographic or environmental gradients if we aim to extend our understanding of adaptive evolution in plants.

In this study, we have shown that the genetic co-variation between flowering time and seed dormancy is geographically fine-tuned by minimum temperature in Mediterranean *A. thaliana*. However, it remains to be seen to what extent the timing of flower initiation affects the timing of seed germination, *e.g.* maternal environmental effects on seed dormancy and germination patterns, or, conversely, whether the timing of seed germination affects the timing of flower initiation, *e.g.* environmental cues for flowering affecting rosettes from seeds germinated at different times (see Donohue 2009). In any case, our results represent a first step towards the development of spatially explicit models to assess the evolutionary dynamics of life-cycle phenology, as well as the impact of climate change on adaptive variation in *A. thaliana* (see non-spatially explicit recent models for *A. thaliana* in Burghardt *et al.* 2015; Fournier-Level *et al.* 2016). This is important because it has been predicted that Mediterranean climate regions, and in particular the Mediterranean Basin, will be dramatically affected by global climate change before the end of the century, with generalised warming and an increase in duration and severity of the summer drought (Klausmeyer & Shaw 2009; Gómez-Navarro *et al.* 2010; Jacobeit *et al.* 2014). In this context, our results suggest that the phenology exhibited by southwest Iberian *A. thaliana*, determined by early flowering and deep seed dormancy, might become the most common life cycle for this annual plant in the climate scenarios forecasted for this region.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article:

Figure S1. (A) Relationship between minimum temperature and the correlation coefficients between DSDS50 and minimum temperature. (B) Relationship between minimum temperature and the correlation coefficients between FT and minimum temperature. Filled dots indicated significant correlation coefficients based on Dutilleul's modified *t*-test. The major turning point of each panel was detected at 7.4 and 6.9 °C for DSDS50 and FT, respectively. In all three cases, values were obtained from temperature sliding window analyses and ranked along the minimum temperature gradient.

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